CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The distance formula

1. The centre of a circle is $(x-7, 2x)$. The circle passes through $(-9, 11)$ and its radius is $5\sqrt{2}$ units. Find the possible values of x. CBSE 2026, 2 marks

SOLUTION

The distance from the centre to $(-9, 11)$ equals the radius.

$(x-7+9)^2 + (2x-11)^2 = (5\sqrt{2})^2$

$x^2 + 4x + 4 + 4x^2 - 44x + 121 = 50$, so $5x^2 - 40x + 75 = 0$

$x^2 - 8x + 15 = 0$, so $(x-3)(x-5) = 0$

Answer: $x = 3$ **or** $x = 5$

2. A circle has its centre at $(2a, a-7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a. CBSE 2025, 2 marks

SOLUTION

Radius $= \dfrac{10\sqrt{2}}{2} = 5\sqrt{2}$ units, so the distance from the centre to $(11, -9)$ is $5\sqrt{2}$.

$(2a-11)^2 + (a-7+9)^2 = (5\sqrt{2})^2$

$4a^2 - 44a + 121 + a^2 + 4a + 4 = 50$, so $5a^2 - 40a + 75 = 0$

$a^2 - 8a + 15 = 0$, so $(a-3)(a-5) = 0$

Answer: $a = 3$ **or** $a = 5$

3. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD. (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units CBSE 2024, 1 mark

SOLUTION

D is the mid-point of BC: $D = \left(\dfrac{6+0}{2}, \dfrac{4+0}{2}\right) = (3, 2)$

$AD = \sqrt{(5-3)^2 + (-6-2)^2} = \sqrt{4+64} = \sqrt{68}$

Answer: (a)

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4. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x. (a) $12, -18$ (b) $-12, 18$ (c) $18, 5$ (d) $-9, -12$ CBSE 2024, 1 mark

SOLUTION

Both points have the same y-coordinate, so the distance is $|x - 3| = 15$.

$x - 3 = 15$ or $x - 3 = -15$, so $x = 18$ or $x = -12$.

Answer: (b)

5. Find the distance between $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$. (a) $2\sqrt{10}$ units (b) $4\sqrt{10}$ units (c) $2\sqrt{20}$ units (d) 0 CBSE 2023, 1 mark

SOLUTION

Distance $= \sqrt{(-2\sqrt{5} - 0)^2 + (0 - 2\sqrt{5})^2} = \sqrt{20 + 20} = \sqrt{40} = 2\sqrt{10}$ units

Answer: (a)

6. Two vertices of an equilateral triangle are $(-5, 3)$ and $(5, 3)$, and the origin lies inside the triangle. Find the third vertex. Take $\sqrt{3} = 1.7$. CBSE 2023, 3 marks

SOLUTION

The side joining the two given vertices has length 10, and its mid-point is $(0, 3)$.

The third vertex is equidistant from the two given points, so it lies on the y-axis. Let it be $(0, y)$.

$(0 - 5)^2 + (y - 3)^2 = 10^2$, so $(y - 3)^2 = 75$ and $y = 3 \pm 5\sqrt{3} = 3 \pm 8.5$

$y = 11.5$ puts the whole triangle above the line $y = 3$, so the origin would be outside it. Hence $y = -5.5$.

Answer: $(0, -5.5)$

7. Assertion (A): $P(0, 2)$ is the point where the line $3x + 2y = 4$ meets the y-axis. Reason (R): The distance of $P(0, 2)$ from the x-axis is 2 units. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2023, 1 mark

SOLUTION

On the y-axis $x = 0$: $2y = 4$, so $y = 2$. The line meets the y-axis at $(0, 2)$, so A is true.

The distance of a point from the x-axis is its y-coordinate, here 2 units, so R is true. But R does not tell us why P lies on the line.

Answer: (b)

8. Find the point on the y-axis that is equidistant from $(5, -2)$ and $(-3, 2)$. CBSE 2019, 3 marks

SOLUTION

Let the point be $P(0, y)$. Equate the squares of the two distances.

$(0 - 5)^2 + (y + 2)^2 = (0 + 3)^2 + (y - 2)^2$

$25 + y^2 + 4y + 4 = 9 + y^2 - 4y + 4$, so $8y = -16$ and $y = -2$.

Answer: $(0, -2)$

9. Write the distance of the point $P(x, y)$ from the origin. CBSE 2018, 1 mark

SOLUTION

Distance $= \sqrt{(x-0)^2 + (y-0)^2}$

Answer: $\sqrt{x^2 + y^2}$ **units**

10. Show that $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle. CBSE 2016, 2 marks

SOLUTION

Call the points $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$.

$AB^2 = 3^2 + 4^2 = 25$, $AC^2 = (-4)^2 + 3^2 = 25$ and $BC^2 = (-7)^2 + (-1)^2 = 50$

$AB = AC$, so the triangle is isosceles. $AB^2 + AC^2 = 50 = BC^2$, so by the converse of the Pythagoras theorem $\angle A = 90°$.

So the triangle is right-angled and isosceles. Proved.

11. The point $P(x, y)$ is equidistant from $A(a + b, b - a)$ and $B(a - b, a + b)$. Prove that $bx = ay$. CBSE 2016, 3 marks

SOLUTION

$PA^2 = PB^2$: $(x - a - b)^2 + (y - b + a)^2 = (x - a + b)^2 + (y - a - b)^2$

Use $p^2 - q^2 = (p - q)(p + q)$ on the x terms: $(x - a - b)^2 - (x - a + b)^2 = (-2b)(2x - 2a) = -4bx + 4ab$

On the y terms: $(y - b + a)^2 - (y - a - b)^2 = (2a)(2y - 2b) = 4ay - 4ab$

Adding the two results gives 0: $-4bx + 4ab + 4ay - 4ab = 0$, so $4ay = 4bx$.

So $bx = ay$. Proved.

2 The section formula

12. Where does the mid-point of the segment joining $(5, -4)$ and $(6, 4)$ lie? (a) on the x-axis (b) on the y-axis (c) at the origin (d) on neither axis CBSE 2026, 1 mark

SOLUTION

Mid-point $= \left(\frac{5+6}{2}, \frac{-4+4}{2}\right) = \left(\frac{11}{2}, 0\right)$

Its y-coordinate is 0 and its x-coordinate is not 0.

Answer: (a)

13. In what ratio does the x-axis divide the segment joining $(-6, 5)$ and $(-4, -1)$? Find the point where they meet. CBSE 2026, 3 marks

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-4k-6}{k+1}, \frac{-k+5}{k+1}\right)$.

On the x-axis the y-coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$x = \frac{-4(5)-6}{5+1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio $5 : 1$; the point is $\left(-\frac{13}{3}, 0\right)$

14. Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? CBSE 2025, 1 mark

SOLUTION

$$\text{Mid-point} = \left(\frac{-4+4}{2}, \frac{5+6}{2}\right) = \left(0, \frac{11}{2}\right)$$

Its x-coordinate is 0.

Answer: on the y-axis

15. In what ratio does the y-axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. CBSE 2025, 3 marks

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-k+5}{k+1}, \frac{-4k-6}{k+1}\right)$.

On the y-axis the x-coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$y = \frac{-4(5)-6}{5+1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio $5 : 1$; the point is $\left(0, -\frac{13}{3}\right)$

16. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ CBSE 2024, 1 mark

SOLUTION

The centre is the mid-point of the diameter. Let the other end be (x, y).

$\frac{6+x}{2} = 2$ gives $x = -2$, and $\frac{0+y}{2} = 0$ gives $y = 0$.

Answer: (c)

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17. The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. CBSE 2024, 3 marks

SOLUTION

Let the ratio be $k : 1$. Then $\frac{2k+1}{k+1} = \frac{8}{5}$.

$10k + 5 = 8k + 8$, so $k = \frac{3}{2}$. The ratio is $3 : 2$.

$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$

Answer: ratio $3 : 2$ and $y = \frac{13}{5}$

18. $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P, Q, R and S are the mid-points of AB, BC, CD and DA. Show that the diagonals of $PQRS$ bisect each other. CBSE 2024, 3 marks

SOLUTION

$P = \left(-1, \frac{5}{2}\right)$, $Q = (1, 6)$, $R = \left(3, \frac{5}{2}\right)$, $S = (1, -1)$

Mid-point of diagonal $PR = \left(\frac{-1+3}{2}, \frac{5/2 + 5/2}{2}\right) = \left(1, \frac{5}{2}\right)$

Mid-point of diagonal $QS = \left(\frac{1+1}{2}, \frac{6-1}{2}\right) = \left(1, \frac{5}{2}\right)$

The diagonals PR and QS have the same mid-point, so they bisect each other.

19. In what ratio does the x-axis divide the segment joining $A(3, 6)$ and $B(-12, -3)$? (a) $1 : 2$ (b) $1 : 4$ (c) $4 : 1$ (d) $2 : 1$ CBSE 2023, 1 mark

SOLUTION

Let the ratio be $k : 1$. On the x-axis the y-coordinate is 0: $\frac{-3k+6}{k+1} = 0$, so $k = 2$.

Answer: (d)

20. AB is a diameter of a circle with centre $(2, -3)$, and B is the point $(1, 4)$. Find the coordinates of A. CBSE 2019, 1 mark

SOLUTION

The centre is the mid-point of the diameter. Let $A = (x, y)$.

$\frac{x+1}{2} = 2$ gives $x = 3$, and $\frac{y+4}{2} = -3$ gives $y = -10$.

Answer: $A(3, -10)$

21. In what ratio does the x-axis divide the segment joining $(1, -3)$ and $(4, 5)$? Find the point of division. CBSE 2019, 2 marks

SOLUTION

Let the ratio be $k : 1$. On the x-axis the y-coordinate is 0: $\frac{5k - 3}{k + 1} = 0$, so $k = \frac{3}{5}$.

The ratio is $3 : 5$. Then $x = \frac{3 \times 4 + 5 \times 1}{3 + 5} = \frac{17}{8}$.

Answer: ratio $3 : 5$; the point is $\left(\frac{17}{8}, 0\right)$

22. The segment joining $A(2, 1)$ and $B(5, -8)$ is trisected at P and Q, with P nearer to A. P also lies on the line $2x - y + k = 0$. Find k. CBSE 2019, 3 marks

SOLUTION

P divides AB in the ratio $1 : 2$.

$P = \left(\frac{1 \times 5 + 2 \times 2}{3}, \frac{1 \times (-8) + 2 \times 1}{3}\right) = (3, -2)$

P lies on the line: $2(3) - (-2) + k = 0$, so $8 + k = 0$.

Answer: $k = -8$

23. In what ratio does $P(4, m)$ divide the segment joining $A(2, 3)$ and $B(6, -3)$? Find m. CBSE 2018, 2 marks

SOLUTION

Let the ratio be $k : 1$. Compare x-coordinates: $\frac{6k + 2}{k + 1} = 4$, so $6k + 2 = 4k + 4$ and $k = 1$.

The ratio is $1 : 1$, so P is the mid-point: $m = \frac{3 + (-3)}{2} = 0$.

Answer: ratio $1 : 1$; $m = 0$

24. $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$. Find a and b, and the lengths of its sides. CBSE 2018, 3 marks

SOLUTION

The diagonals of a parallelogram bisect each other, so AC and BD have the same mid-point.

$\frac{-2 + 4}{2} = \frac{a + 1}{2}$ gives $a = 1$, and $\frac{1 + b}{2} = \frac{0 + 2}{2}$ gives $b = 1$.

$AB = \sqrt{(1 + 2)^2 + (0 - 1)^2} = \sqrt{10}$ and $BC = \sqrt{(4 - 1)^2 + (1 - 0)^2} = \sqrt{10}$

Opposite sides are equal, so $CD = \sqrt{10}$ and $DA = \sqrt{10}$.

Answer: $a = 1$, $b = 1$; each side is $\sqrt{10}$ units

25. P and Q are the points of trisection of the segment joining $A(2,-2)$ and $B(-7,4)$, with P nearer to A. Find P and Q. CBSE 2016, 2 marks

SOLUTION

P divides AB in the ratio $1:2$: $P = \left(\frac{-7+4}{3}, \frac{4-4}{3}\right) = (-1, 0)$

Q divides AB in the ratio $2:1$: $Q = \left(\frac{-14+2}{3}, \frac{8-2}{3}\right) = (-4, 2)$

Answer: $P(-1,0)$ **and** $Q(-4,2)$

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